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ORIGINAL PAPER

Wave group focusing in the ocean: estimations using crest velocities and a Gaussian linear model

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Abstract

Wave group focusing gives rise to the formation of large gravity waves at the surface of the ocean, some of which are called rogue waves and represent a natural hazard for ships and offshore platforms. For safety purposes, it is crucial to predict when and where these large waves will appear and how large they will be. This work focuses on crest velocities, a quantity that is relatively easy to extract from sea surface elevation fields. It is shown that there is a direct link between crest velocity gradient and wave group linear dispersive focusing. Studying analytically the focusing of one-dimensional Gaussian wave packets under linear evolution makes it possible to derive estimates of quantities at focus, based only on crest velocity measurements. In this way, the focusing time, focusing size and focusing amplitude (relative to instantaneous amplitude) of an isolated Gaussian wave packet can be estimated. Our work is also applicable to second-order non-linear waves. Limitations due to higher-order non-linear effects are studied in numerical simulations of the non-linear Schrödinger equation.

Keywords Ocean wave · Rogue wave · Wave group · Wave packet · Crest velocity · Focusing · Schrödinger equation

1 Introduction

Human and material losses caused by large amplitude ocean surface waves are significant, as reported by Kharif and Pelinovsky (2003). The term “rogue wave” refers to waves with a crest-to-trough wave height larger than $2H_s$, where H_s is the significant wave height and

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is defined as four times the standard deviation of the sea surface elevation (Dysthe et al. 2008). The threshold of $2H_s$ is arbitrary, and in the following, "rogue wave" might simply refer to waves with a large height compared to local statistics. Many rogue waves have been measured in the ocean, including the famous 26-m-high Draupner wave (Haver 2004). Even in moderate sea states, the occurrence of rogue waves can be problematic for operations (e.g. cable layout from a ship) or for transferring staff from a ship to an offshore platform or wind turbine. The harmful potential of rogue waves is a natural motivation for the forecast of their position, amplitude and spatial width over time, based on real-time measurements available from instruments that can be mounted on ships or offshore platforms. The short-term forecast of high-amplitude ocean waves must neither be confused with tsunami forecast (Titov et al. 2005) nor sea state forecast (Niclasen et al. 2010) nor short-term forecast of sea surface elevation for wave energy converters (Fusco and Ringwood 2010).

Many physical processes could be involved in the formation and growth of oceanic rogue waves (Adcock and Taylor 2014). In this paper, we focus on the formation of large surface gravity waves in deep water due to dispersive focusing, a process that we briefly describe here. In deep water, the sea surface elevation can be approximated by an infinite sum of plane waves propagating at a velocity given by the dispersion relation of ocean surface gravity waves (Longuet-Higgins 1657). This approximation is based on the linearization of the physical equations at the ocean surface, assuming small steepness of the sea surface elevation. Dispersion causes waves of different wavelengths to propagate at different velocities, which creates mixing. When this mixing results in the constructive interference of a large number of waves (superposition of their crests, or troughs), this leads to a wave of large amplitude, and we call this process dispersive focusing. Here, we use a linear description of long-crested (one-dimensional in space) ocean waves, and we tackle the issue of forecasting the dispersive focusing of ocean wave packets based on crest velocities. Thus, the proposed methodology is not limited to waves exceeding the rogue threshold of $2H_s$, but aims at forecasting the maximum amplitude of any wave packet under dispersive focusing.

Other processes can lead to the focusing of ocean waves, such as wave current interactions (Peregrine 1976; Quilfen et al. 2018). Also, the focusing of ocean waves can increase wave steepness, leading to second-order non-linear effects that act on the shape and height of waves (Forristall 2000), or even higher-order non-linearities potentially leading to the instability of Benjamin and Feir (1967). However, we ignore wave current interactions and first investigate linear dispersive focusing. The latter appears to be a necessary first step in the formation of the largest ocean surface gravity waves (Fedele 2008) and is thought to be the main mechanism driving the formation and evolution of real-life rogue waves, along with second-order non-linearities (Fedele et al. 2016). Although designed for linear waves, our work is applicable to second-order non-linear waves, and we also examine the limitations of our study due to higher-order non-linear effects.

A review of the possible strategies for rogue wave forecasting in the absence of currents and based on space-time measurements of the sea surface elevation only can be found in Slunyaev (2017). For a review and comparison of numerical methods for deterministic short-term prediction of water wave fields, the reader is referred to Klein et al. (2020), which also lists commercial decision support systems based on X-band radar acquisition. However, the forecast of high-amplitude waves does not require prediction of the whole wave field, and forecast methods may be targeted only at high-amplitude waves. Slunyaev et al. (2005) developed the idea that rogue wave formation could be inferred from energy flux convergence. A predictive scheme based on such a principle was developed by Guo et al. (2017). However, the method laid out in Guo et al. (2017) requires knowledge about the field of fluid velocity in the whole water column, a quantity that is difficult to measure in real time. Slunyaev (2017) proposed that the energy transfer could be inferred from the local group velocity, an idea which is closely related to what we propose here.

The envelope of the wave field is not well defined when the wave spectrum is not narrow enough. For this reason, our method relies on the velocities of crests and troughs, which are theoretically accessible from fields of sea surface elevation. Today, however, in situ ocean wave measurement technology needs to improve in order to achieve the forecasting of large ocean waves a few minutes in advance, independently of the method that is to be used. The combination of X-band radar images (Borge et al. 2013) with stereo video cameras (Benetazzo et al. 2018; Filipot et al. 2019) is promising, and motivated us to focus on crest velocities. For the same applications, the sea surface reconstruction using lidar illumination is another encouraging alternative (Nouguier et al. 2013).

Figure 1 shows the basic principle of the proposed methodology. A wave packet under linear evolution in a dispersive medium (where the phase velocity is frequency-dependent) is either in the stage of focusing (a, d), is focused (b, e), or is in the stage of defocusing (c, f), and those three stages always take place in this order. When the wave packet is focused, the velocities of the crests and troughs (maxima and minima) are nearly homogeneous (e) and smaller than the phase velocity corresponding to the peak frequency of the Gaussian wave packet (GWP). This slowdown effect is described in the linear wave theory outlined by Fedele (2014). Conversely, when the wave packet is not focused, crest and trough velocities are neither uniform in space nor constant in time (d, f): there is a strong gradient of crest velocity (or, to be more precise, a gradient of specular velocity, see Sects. 2, 2.3). The

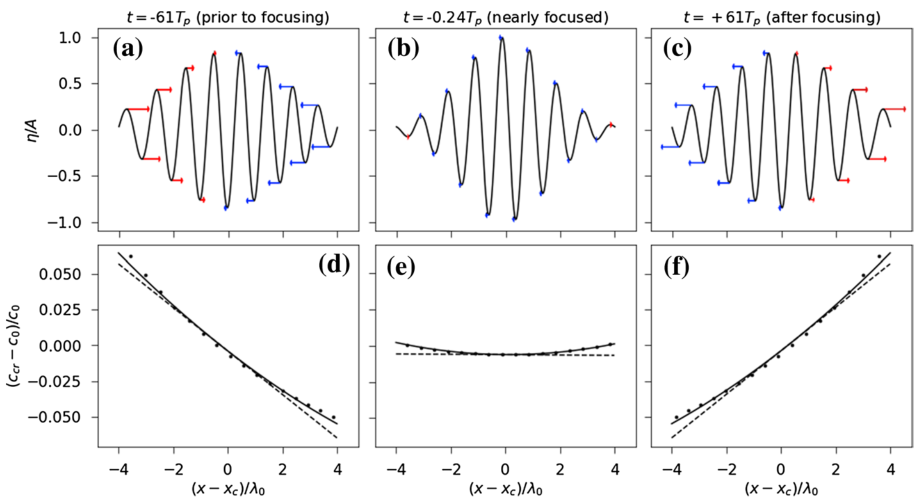

Fig. 1 Crest and trough velocities of a Gaussian wave packet (GWP) at different times under linear evolution. The GWP is moving from left to right, and we assume the dispersion relation of deep-water gravity waves. The peak phase velocity is noted c_0. Extracted from the numerical experiments described in Sect. 3, with $L_f k_0 = 10$. Abscissa: position relative to the centre of the GWP normalized by the peak wavelength $\lambda_0 = 2\pi/k_0$. Black curves in **a**–**c** are a numerical approximation of η_G calculated using Eq. (1). Arrows in **a**–**c** have lengths proportional to $c_{cr} - c_0$, red when $c_{cr} > c_0$ and blue when $c_{cr} < c_0$. Black dashed lines in $(d-f)$ are a first-order approximation of the analytical prediction c^{a}_{sp}. Black curves in **d**–**f** are second-order approximation. Black dots in **d**–**f** are crest velocities c_{cr} extracted from numerical experiment. Shown are the situations prior to focusing (**a**, **d**), nearly focused (**b**, **e**) and after focusing (**c**, **f**)

sign of this gradient reveals the focusing or defocusing nature of the wave packet, while its absolute value is proportional to the growth rate of the wave packet amplitude (for narrow-banded linear wave packets). By tracking the evolution of this gradient over time, it is possible to predict the time at which the GWP is focused, the size of the GWP at focus, and the ratio between the amplitude at focus and the instantaneous amplitude.

Theoretical background and hypotheses are outlined in Sect. 2, with details of derivations given in "Appendix." The method is tested in numerical experiments of linear waves in Sect. 3. In Sect. 4, we investigate the applicability of the method to weakly non-linear wave packets using numerical experiments of the non-linear Schrödinger equation.

2 Theory for linear Gaussian wave packets

2.1 Definitions and hypotheses

We refer to Gaussian wave packets (GWPs) as one-dimensional sea surface elevation profiles η_G such that the modulus of the space Fourier transform of η_G is a Gaussian function of wave number k. Such packets were widely used to generate water waves in laboratory experiments (Clauss and Bergmann 1986) and to study theoretically and numerically the dispersive formation of large amplitude waves (for instance, see Pelinovsky et al. 2011). The Gaussian shape assumption allows us to carry detailed calculations for the crest velocities. We can write η_G as the real part of its complex analytical representation Ψ_G with

$$\Psi_G(x,t) = \frac{A_\mathrm{f} L_\mathrm{f}}{\sqrt{2\pi}} \int \exp\left\{ -\frac{L_\mathrm{f}^2 (k-k_0)^2}{2} + i(kx - \Omega(k)t + \Phi(k)) \right\} \mathrm{d}k, \tag{1}$$

where x is position, t is time, A_f is the amplitude of the GWP at focus, L_f is the spatial extent of the GWP at focus (also further referred to as "group size"), k_0 is the peak wave number of the GWP, $\Omega(k)$ is the angular frequency given by the dispersion relation and $\Phi(k)$ is the phase of the GWP. All quantities are real unless otherwise stated.

We make the narrow-band approximation that $L_\mathrm{f} k_0$ is large, which induces relative errors that are all proportional to a negative power of $L_\mathrm{f} k_0$. The effects of surface current, wind input and wave dissipation are ignored. Non-linearities are first neglected, but their effect is studied in the last section.

2.2 Approximation of Ψ_G

Assuming that $(L_\mathrm{f} k_0)^{\frac{1}{2}} >> 1$, and that Ω and Φ are smooth enough (see "Appendix"), and choosing the origins of the x-axis and t-axis as suggested at the end of "Appendix," we have $\Psi_G \approx \Psi_G^\mathrm{a}$ and

$$\Psi_G^\mathrm{a}(x,t) = A(t) \exp\left\{ \frac{-(x - x_c(t))^2}{2L(t)^2} + i\,(k_0 x - \omega_0 t + \phi(x,t)) \right\}, \tag{2}$$

where the a superscript stands for "approximation." In Eq. (2), $\omega_0 = \Omega(k_0)$, and $x_c = c_g t$ is the centre of the GWP, and we have

$$\frac{A(t)}{A_\mathrm{f}} = \left(1 + (t/\tau)^2\right)^{-1/4}, \quad \frac{L(t)}{L_\mathrm{f}} = \left(1 + (t/\tau)^2\right)^{1/2}, \tag{3}$$

$$\phi(x,t) = \phi_0 - \arctan(t/2\tau) + (t/2\tau)\frac{(x-x_c)^2}{L(t)^2}, \tag{4}$$

where $\tau = k_0 L_\mathrm{f}^2 / c^*$ is the group contraction timescale (possibly negative), $c_g = \frac{\mathrm{d}\Omega}{\mathrm{d}k}(k_0)$ and $c^* = k_0 \frac{\mathrm{d}^2\Omega}{\mathrm{d}k^2}(k_0)$. From our hypotheses, we have $|\tau| \ll \omega_0^{-1}$ so that $A(t)$, $L(t)$ and $\phi(t)$ are slowly varying compared to the carrier plane wave $e^{i(k_0 x - \omega_0 t)}$. See Fig. 2 for an illustration of these quantities. The approximation $\Psi_G \approx \Psi_G^\mathrm{a}$ induces a relative error of $O\left(L_\mathrm{f} k_0\right)^{-4}$.

The expression of Ψ_G^a was given by Kinsman (1965) for the case $\Phi(k) = 0$. In Eq. (2), we generalize to the case where $\Phi(k)$ is a slowly varying function of k.

2.3 Crest velocity approximation

In one dimension, we can follow the trajectory of a crest $x_\mathrm{cr}(t)$ by imposing

$$\partial_{xx}\eta\left(x_\mathrm{cr}(t_0), 0\right) < 0,\ \forall t > t_0,\ \partial_x \eta\left(x_\mathrm{cr}(t), t\right) = 0.$$

This implies that $c_\mathrm{cr}(t) = c_\mathrm{sp}(x_\mathrm{cr}(t), t)$ where $c_\mathrm{sp} = -\frac{\partial_{xt}\eta}{\partial_{xx}\eta}$ is the velocity of points of constant slope called specular points (Longuet-Higgins 1657). Specular points would be indicated to a distant observer as the points where light was reflected from a distant source. The velocity of specular points can be defined at every point of the surface and contains the crest velocities. For identical reasons, trough velocities can also be calculated using the velocity of specular points. Using Eq. (2), we can express c_sp^a (an approximate form of c_sp based on Ψ_G^a) as a function of $X \equiv k_0(x - x_c)$ and $T \equiv t/\tau$:

$$c_\mathrm{sp}^\mathrm{a}(X,T) = \frac{C_1(T)X^4 + C_2(T)X^3 + C_3(T)X^2 + C_4(T)X + C_5(T)}{C_6(T)X^3 + C_7(T)X^2 + C_8(T)X + C_9(T)}$$

with the $C_i(T)$ being time-dependent coefficients which all have terms proportional to the small parameter $(L(t)k_0)^{-2}$. For the sake of brevity, we do not give the expressions of those coefficients, but only the first terms of the Taylor expansion of c_sp^a with respect to $(L(t)k_0)^{-2}$. We also assume the dispersion relation of deep-water gravity waves (i.e. $\Omega \propto \sqrt{k}$)

$$\frac{c_\mathrm{sp}^\mathrm{a}(X,T)}{c_0} = 1 + \frac{5}{8}\frac{[-1 - 0.8XT]}{(L(t)k_0)^2} + \frac{5}{8}\frac{\left[1 + XT + \left(0.2 + T^2\right)X^2\right]}{(L(t)k_0)^4} + \mathcal{O}\left(\frac{1}{(L(t)k_0)^6}\right), \tag{5}$$

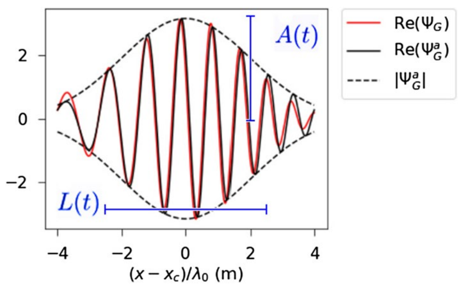

Fig. 2 A Gaussian wave packet a few periods before focusing, with related notations. The approximation Φ_G^a and Φ_G are almost identical, and the largest discrepancies are found at the tails of the GWP. This GWP has parameter $L_\mathrm{f} k_0 = 5$ and $A_\mathrm{f} = 5$

where $c_0 = \omega_0/k_0$. Neglected higher-order terms are of $\mathcal{O}\big((Lk_0)^{-6}\big)$ and can only become large when $|x - x_c| \gg L(t)$ or $|t/\tau| \gg 1$, cases where the height of the waves is negligible. As pointed out by one of the reviewers and as mentioned in "Appendix," the fourth-order terms in this expression have been derived from Ψ_G^a which is a second-order approximation of Ψ_G. For this reason, they cannot be trusted in theory. However, the numerical experiments shown in this paper suggest that they give a reasonable approximation of c_{sp}.

At zeroth order, $c_{\mathrm{cr}}^{\mathrm{a}}$ equals c_0, which corresponds to the monochromatic case. At first order, it is a time-dependent linear function of X (see the dashed lines of Fig. 1d–f). At second order, it is a time-dependent second-order polynomial in X (see the full curves of Fig. 1d–f). Good agreement is found between $c_{\mathrm{cr}}^{\mathrm{a}}$ which was derived from the approximate Ψ_G^{a}, and c_{cr} which is associated with Ψ_G (see the dots of Fig. 1d–f). Other experiments with lower values of k_0L_f also show a good agreement between $c_{\mathrm{cr}}^{\mathrm{a}}$ from Eq. (5) and c_{cr} (not shown).

Note that Eq. (5) is valid only for specular points of slope zero $\partial_x\eta = 0$, which is the case for crests and troughs. The general expression for c_{sp}^{a} is more complicated than Eq. (5), but by abuse of notation and for our purpose c_{sp}^{a} denotes the right-hand side of Eq. (5), which is a Eulerian quantity, while c_{cr} is Lagrangian. In particular, this allows us to compute derivatives of the right-hand side of Eq. (5).

3 Numerical experiments of linear waves

3.1 Numerical set-up

We performed simulations based on a numerical approximation of the integral in Eq. (1). All parameters were fixed except $L_\mathrm{f}k_0$. We assumed the dispersion relation of deep-water gravity waves $\Omega(k) = \sqrt{gk}$ where g is the acceleration of gravity at the earth's surface, set to 9.81 m s^{-2}. Φ was set to be a constant without loss of generality up to a change of reference frame, so that the GWP is focused at $t = 0$. Setting $A_\mathrm{f} = 5$ m and $k_0 = 0.02$ rad m^{-1} gives $c_0 \approx 22.15$ m s^{-1} and $T_p = 2\pi/\sqrt{gk_0} \approx 14$ s. Our choice for k_0 corresponds to typical dominant surface gravity waves encountered in the ocean. We did numerical experiments with three different values for $L_\mathrm{f}k_0$, either $L_\mathrm{f}k_0 = 2.36$ or $L_\mathrm{f}k_0 = 5$ or $L_\mathrm{f}k_0 = 10$.

In all experiments, the integral was approximated using 150 wave numbers, ranging from $k_{\mathrm{min}} = 0.001$ rad m^{-1} to $k_{\mathrm{max}} = 0.05$ rad m^{-1} when $L_\mathrm{f}k_0 = 2.36$; from $k_{\mathrm{min}} = 0.005$ rad m^{-1} to $k_{\mathrm{max}} = 0.037$ rad m^{-1} when $L_\mathrm{f}k_0 = 5$; and from 0.012 rad m^{-1} to 0.028 rad m^{-1} when $L_\mathrm{f}k_0 = 10$. Further refinement in the wave number discretization did not yield significant changes in the results. The integral defining $\Psi_G(x,t)$ was evaluated over a space-time grid with constant steps d$x = 3.14$ mm and d$t = 1.17$ s when $L_\mathrm{f}k_0 = 2.36$; d$x = 2.55$ mm and d$t = 1.36$ s when $L_\mathrm{f}k_0 = 5$; and d$x = 3.37$ mm and d$t = 1.56$ s when $L_\mathrm{f}k_0 = 10$. In all experiments, $(x - x_c)$ varies from $-2\lambda_0$ to $+2\lambda_0$. t varies from $-4|\tau| \approx 14\,T_p$ to 0 when $L_\mathrm{f}k_0 = 2.36$; from $-2|\tau| \approx 32\,T_p$ to 0 when $L_\mathrm{f}k_0 = 5$; and from $-|\tau| \approx 64\,T_p$ to 0 when $L_\mathrm{f}k_0 = 10$. The time step was chosen to be comparable with X-band radar typical measurement frequency, which are on the order of 1 Hz. Note that high spatial resolutions allow the precise measurement of small crest velocity variations (see the vertical axis of Fig. 1d–f) for convenience in testing the method. In practice, time-averaging techniques could be used to lower-spatial-resolution constraints. Each experiment was run a thousand times for different values of the constant phase Φ, picked randomly between $-\pi$ and π.

3.2 The choice of $L_f k_0$

The typical spectral shape of individual wave packets in the ocean is unknown, and thus, our justification for the choices of values for $L_f k_0$ will only be partial. Isolated wave packets only exist conceptually, and in the ocean, one wave packet is always at least partially merged with other neighbouring packets. Ocean wave measurements allow for the reliable estimation of the spectrum of a sample of the sea surface. Such a sample is made of many packets merged together. We will call such a spectrum a "total spectrum" to make a clear distinction with the spectrum of one individual packet. To our knowledge, there is no theory which gives the conditional probability that a measured wave packet has a given spectrum (corresponding to A_f, L_f and k_0 in our case), assuming that the total spectrum is known. Longuet-Higgins (1984) links the temporal width of wave packets (linked to our $L(t)k_0$) with the width of the total spectrum ν. But the probability of measuring a packet with an instantaneous size $L(t)k_0$ is different from the probability for a given packet to have a focusing size $L_f k_0$. However, a wave packet that is far from focusing will have a large spatial width, but a small amplitude, and it will thus have a lower probability of being measured at sea as it would be merged with other packets of greater amplitude.

Cousins and Sapsis (2015b) performed numerical simulations of ocean waves with Gaussian spectra and randomly drawn phases, coupled with an algorithm for the detection of the spatial width and amplitude of individual wave packets. From those simulations, the authors were able to compute probability distributions for a given wave packet to be observed at any point in time, with a given spatial width and amplitude. In those numerical simulations, the packets observed with an amplitude larger than the rogue wave threshold $A(t) > H_s$ had a spatial width $L(t)k_0$ close to the inverse of the width of the total spectrum. As those packets had large amplitudes, it is likely that they were focused or close to focusing. This seems to indicate that for a given spectrum, the packets which have a high amplitude are likely to have a spectral width comparable to that of the total spectrum.

The case $L_f k_0 = 2.36$ corresponds to a spectral width of 0.15, $L_f k_0 = 5$ corresponds to 0.071, and $L_f k_0 = 10$ corresponds to 0.035. Recall that we use the definition of spectral width through moments of the temporal frequency: $\nu^2 = m_0 m_2 / m_1^2 - 1$ and $m_i \propto \int S(f) f^i \mathrm{d}f$ for all i with f the temporal frequency and $S(f)$ the frequency spectrum, i.e. the density of energy per frequency. In the ocean, the most dangerous waves are due to large values of H_s, associated with storm seas. In these sea states, the largest individual waves are the so-called dominant waves that contain the energy of a narrow spectral band around the peak frequency, f_p. Banner et al. (2000) suggested that this spectral band ranges from $0.7f_p$ to $1.3f_p$. Computing ν over this spectral band for storm wave spectra computed at a deep-water point in the North East Atlantic ([−4.5° W, 46° N]) during the entire 2013–2014 winter and using the WaveWatchIII spectral wave model (Tolman et al 2009) leads to values lying between 0.05 and 0.15. We also note that the directional spreading for this particular spectral band is generally limited (about 20° using the same dataset). Our numerical experiments thus show how our method should behave for dominant waves in typical storm conditions.

3.3 Validity of the narrow-band approximation for c_{sp}^{a}

We know from our analytical derivations based on Ψ_G^{a} that the specular velocity gradient at the centre of the GWP contains information on the focusing properties of the packet, and we also know that this gradient is almost homogeneous in the spatial dimension x.

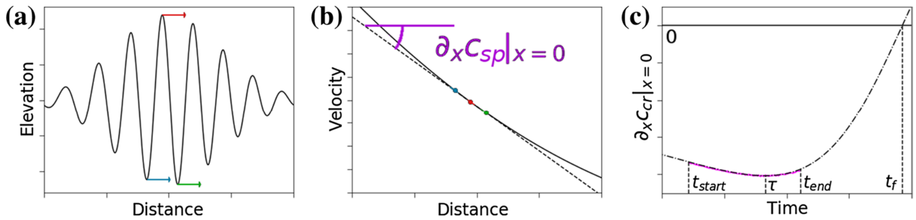

Fig. 3 Step-by-step methodology. **a** Crest detection and velocity estimation. **b** Velocity gradient estimation. **c** Velocity gradient estimation over time, fitted and extrapolated

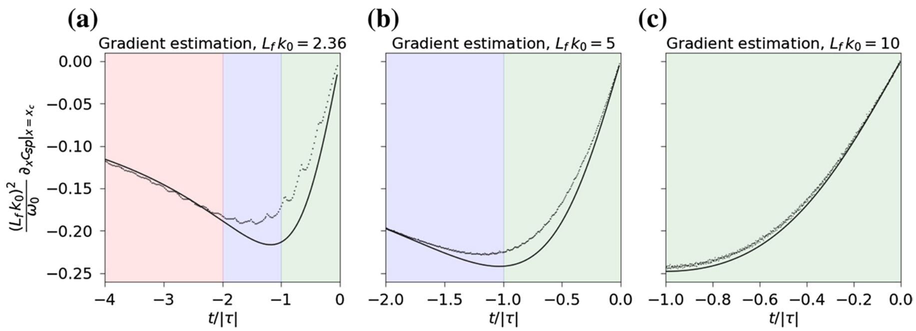

Fig. 4 Normalized crest velocity gradient at the centre of the GWP. The full curves correspond to the second-order analytical prediction of $\partial_x c^{a}_{cr}|_{x=x_c}$ from Eq. (6). The dots are approximations of $\partial_x c_{cr}$ from three crest/trough velocities around $x = x_c$ from the numerical experiments. Green background is for $-|\tau| \leq t \leq 0$, violet for $-2|\tau| \leq t \leq -|\tau|$ and pink for $-4|\tau| \leq t \leq -2|\tau|$

Thus, we hope to estimate this gradient based on measurements of crest and trough velocities around the centre of the GWP. We start by measuring the crest velocities of the three crests/troughs that are closest to the centre of the GWP and the distances between them, giving us three values of $c_{\mathrm{sp}}(\cdot, t)$ (see Fig. 3a). We chose to use only three values because we want this procedure to be applicable to any wave packet inside a random unidimensional field (meaning one space dimension and one time dimension) of linear surface gravity waves containing many different packets. In this case, the crests and troughs of a given wave packet that are far away from its centre may be mixed up with crests and troughs from other packets. In the case of a random unidimensional field of linear surface gravity waves, our procedure would be equivalent to finding the largest crest or trough in the packet and measuring its velocity and those of is two neighbouring troughs or crests, respectively.

Then, at any time t we make a polynomial fit of second order in space from our three values of $c_{\mathrm{sp}}(\cdot, t)$ (full curve of Fig. 3b), and we keep the linear part of this fit as our approximation for $\partial_x c_{\mathrm{sp}}|_{x=x_c}$ (dashed curve of Fig. 3b).

We can compare the result of this procedure (Fig. 3a, b) to our approximate analytical prediction $\partial_x c^{\mathrm{a}}_{\mathrm{sp}}|_{x=x_c}$ that we obtain from Eq. (5) when neglecting terms of $\mathcal{O}\big((Lk_0)^{-6}\big)$. This is plotted in Fig. 4. In the case where $L_{\mathrm{f}}k_0 = 2.36$, it can be seen that our procedure causes oscillations in the estimation of $\partial_x c_{\mathrm{sp}}|_{x=x_c}$. This is due to the fact that the phase velocity of surface gravity waves is twice as large as their group velocity, and thus, crests move twice as fast as the centre of the wave packet. Thus, every $\Delta t \approx 2\,T_p$, the crest or trough in the front of the GWP that we use for our approximation of $\partial_x c_{\mathrm{sp}}|_{x=x_c}$ is no longer one of the three closest crests or troughs from the centre of the GWP, and it is thus replaced by another trough or crest, respectively, coming from the back of the GWP. These oscillations still exist for larger values of $L_f k_0$, but they appear faster at the timescale $|\tau|$, since we have $|\tau|/T_p = 2(L_f k_0)^2/\pi$ from the definitions in Sect. 2.2. As $L_f k_0$ decreases, our approximations are less valid, especially around $t = -|\tau|$, but the behaviour of our measured $\partial_x c_{sp}|_{x=x_c}$ is still coherent with $\partial_x c^a_{sp}|_{x=x_c}$ and the approximations are relevant when t is far enough from $-|\tau|$. It can be observed that for $L_f k_0 = 5$ and 10, the minimum of our approximation for $\partial_x c^a_{sp}|_{x=x_c}$ is achieved close to $t = -|\tau|$. This is because the term in parentheses in Eq. (6) is close to 1. On the other hand, for $L_f k_0 = 2.36$, we can see that the minimum of $\partial_x c^a_{sp}|_{x=x_c}$ is achieved slightly before $t = -|\tau|$, and is of lower relative amplitude.

3.4 Estimation of focusing time, amplitude and width

Once we have an approximate measure of $\partial_x c_{sp}|_{x=x_c}$ over time between t_{start} and t_{end}, we want to find relevant information related to the focusing of the wave packet, such as t_f, $A_f/A(t_{end})$ and L_f. To do this, we fit our approximate measure of $\partial_x c_{sp}|_{x=x_c}$ over time to a known profile derived from Eq. (5), which has the following form

$$\partial_x c^a_{sp}|_{x=x_c}(t) \approx \frac{2(t-t_f)}{\tau^2 + (t-t_f)^2}\left(1 - \frac{13/\omega_0|\tau|}{1+\left((t-t_f)/\tau\right)^2}\right). \tag{6}$$

The fitting step corresponds to Fig. 3c, and the right-hand side of Eq. (6) is plotted in Fig. 4. We assume that ω_0 is known; thus, the only parameters to be estimated are $|\tau|$ and t_f. These parameters can be read on the graph in Fig. 3c as t_f is at the crossing with the $y = 0$ horizontal line and τ is approximately at the minimum of the curve. Then, using $|\tau|$, t_f and Eq. (3), we can estimate $A_f/A(t_{end})$, and further using ω_0, we estimate $L_f = \sqrt{|\tau|\omega_0}/k_0$. In Fig. 5, we show the results of this procedure for the estimation of t_f/T_p, $A_f/A(t_{end})$ and $L_f k_0$, for our three values of $L_f k_0$ and for $t_{start} = -4|\tau|; -2|\tau|; -|\tau|$ and t_{end} going from t_{start} to $t_f = 0$. When $t_{end} - t_{start}$ is small compared to $|\tau|$, large variations in the values of the fitting parameters are observed. This variability is the consequence of variations in the estimation of $\partial_x c_{sp}|_{x=x_c}$ which happen at the small timescale T_p, such as the oscillations described in Fig. 4. When $t_{end} - t_{start}$ is only a few T_p, those small timescale variations are incorrectly fitted, as if they corresponded to the profile from Eq. (6), which varies at the large timescale $|\tau|$. This causes incorrect estimations of the parameters t_f and τ. Systematic bias can be observed when $L_f k_0 = 5$ and 10, with an overestimation of focusing amplitude and group size, and underestimation of focusing time. This is consistent with the systematic bias observed in Fig. 4. It might be argued that for warning purposes, an error of less than 10% might not be critical, and thus, that the systematic bias observed for $L_f k_0 = 5$ and 10 should not be cause for concern. Estimations in the case where $L_f k_0 = 2.36$ are challenging, because deviations from the hypothesis on which our method relies start to be large. However, our estimated values are still acceptable, which indicates a certain robustness in our method. An interesting fact is that as $L_f k_0$ decreases, relative errors increase, but when coming back to the typical scales of ocean waves, these errors are of comparable amplitude. See, for instance, the green curve in Fig. 5h, with a systematic bias of $\approx 2\%$, which gives an error of $\approx 0.1 k_0^{-1}$ for L_f; similarly, the green curve in Fig. 5i, with a bias of

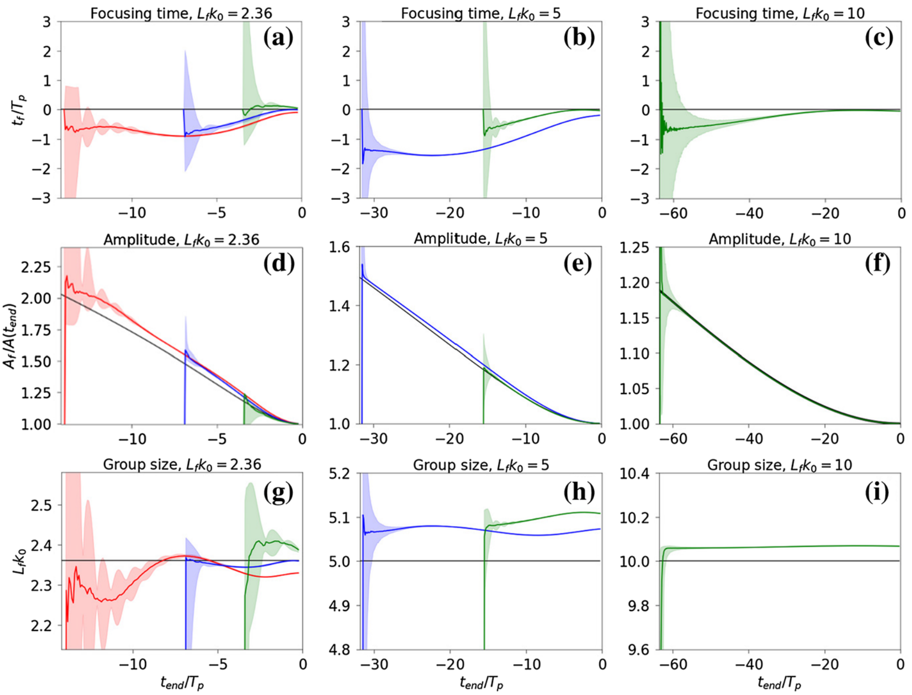

Fig. 5 Estimation of focusing time t_f (**a**–**c**), ratio of focusing amplitude to instantaneous amplitude $A_f/A(t_{end})$ (**d**–**f**), and group focusing size L_f (**g**–**i**), following the methodology described in Sect. 3.4. The objective (truth) is represented in each subplot by a black line. Estimations are based on approximate measurements of $\partial_x c_{sp}|_{x=x_c}$ between t_{start} and t_{end}, with $t_{start} = -4|\tau|$ (red) or $t_{start} = -2|\tau|$ (blue) or $t_{start} = -|\tau|$ (green). Experiments are repeated with different values of the phase Φ drawn randomly between $-\pi$ and π. The medians of predictions are shown in full lines and points between the 2.5 and 97.5 percentiles are shown in light shading. $L_f k_0 = 2.36$ for **a**, **d** and **g**; $L_f k_0 = 5$ for **b**, **e** and **h**; and $L_f k_0 = 10$ for **c**, **f** and **i**. Vertical scales of **g**–**i** are proportional to the value of Lfk_0

$\approx 0.5\%$, gives an error of $\approx 0.05 k_0^{-1}$ for L_f. Estimations of t_f give biases of the order of T_p, while those biases would seem much lower for $L_f k_0 = 10$ if we had plotted estimates of t_f/τ rather than t_f/T_p.

As the time of contraction scales with τ (and thus with L_f^2), focusing properties of large groups (or groups of narrow spectra) can be predicted accurately a long time in advance, but they also require that the group crest velocity gradient be tracked far in advance. We note that it is technically challenging to predict the focusing of a wave packet 60 T_p in advance, because it requires having information on the wave field in a radius of about 10 km (for a typical peak period of 10 s) around the point of interest.

In our experiments, we assumed that ω_0 is known. If ω_0 was unknown, we still could have used the lower-order approximation $\partial_x c_{sp}^a|_{x=x_c}(t) \approx 2(t-t_f)/(\tau^2 + (t-t_f)^2)$, which does not depend on ω_0, and use it to evaluate t_f and $A_f/A(t_{end})$. The impact of such an approximation should not be strong in cases where $L_f k_0 = 5$ and 10, as we showed that including higher-order terms did not yield a significant change in the profile (Fig. 4). Then, however, no estimation of L_f could have been done.

4 Non-linear effects

4.1 General statements

Our work is based on the assumption of linear waves corresponding to infinitely small amplitudes $k_0A_\mathrm{f} \to 0$. Yet, real ocean waves have a finite amplitude and are thus non-linear. We examine here the consequences of non-linear effects on the validity of our method.

4.1.1 Second-order non-linearity

Second-order non-linearity affects the shape and amplitude of ocean waves. It makes crests sharper and troughs shallower, and raises the heights of crests and troughs by a quantity that is proportional to the product of the linear wave amplitude and the wave steepness (Forristall 2000). Second-order non-linear effects modify the wave spectrum, although the latter is still time-independent. These effects modify neither the crest-to-trough wave height nor the position of crests and troughs.

Here, we use crest and trough velocities to predict GWP focusing time, focusing amplitude and focusing width. All the quantities mentioned in the last sentence are identical for linear and second-order non-linear waves. Therefore, our method is as valid for second-order non-linear waves as for linear waves. This is important because, for linear waves, the dispersion relation of surface gravity waves allows for very simple forecasting of the whole sea surface using Fourier transforms. But this forecasting strategy cannot be applied to second-order non-linear waves for which the dispersion relation does not hold. The prevalence of second-order non-linear effects in the ocean thus further highlights the relevance of our method for real ocean waves. Note, however, that, from a practical point of view, the change in wave shape due to second-order non-linearity might make the accurate measurement of trough positions more challenging.

4.1.2 Higher-order non-linearity

Higher-order (third-order and above) non-linear effects induce changes in shape, amplitude and width of ocean wave packets. They make wave packet contraction stronger in the principal direction of propagation and weaker in the orthogonal direction (Adcock et al. 2012). Higher-order non-linearities can create surface gravity waves of very high amplitudes compared to linear waves and have thus been studied extensively in the literature on rogue ocean waves. The spectrum of higher-order non-linear ocean waves is time-dependent, so the focusing time of higher-order non-linear wave packets is different from that of linear ones.

All these elements indicate that our forecasting strategy based on the linear wave hypothesis will fail to accurately forecast the focusing time, amplitude and width of a higher-order non-linear wave packet. However, we expect the quantities we want to predict to be continuous functions of the non-dimensional amplitude k_0A_f which controls the degree of non-linearity of the wave packet. Therefore, our method should still provide interesting results for finite but moderate values of k_0A_f.

Following this preliminary analysis, we investigate the effects of higher-order non-linearities through numerical experiments of the one-dimensional non-linear Schrödinger equation (NLSE).

4.2 Numerical resolution of the NLSE

The one-dimensional NLSE, first described by Zakharov (1968), is a simple model for the evolution of the envelope of narrow-banded non-linear wave groups in deep water. Although it can model non-linearities above the second order, it is only valid for waves of moderate amplitude (called weakly non-linear waves). This is not a problem for us, as our concern is investigating the effects of higher-order linearities of moderate strength on our method. Comparison between solutions of the NLSE and fully non-linear simulations of water waves can be found in Henderson et al. (1999). In the reference frame moving at the group velocity, the NLSE can be written as follows:

$$i\partial_t u = \frac{\omega_0}{8k_0^2}\partial_x^2 u + \frac{1}{2}\omega_0 k_0^2 |u|^2 u, \tag{7}$$

where $u(x, t)$ is the envelope of the wave field at position x and time t, ω_0 is the peak pulsation and k_0 is the peak wave number. In the reference frame moving at the group velocity, the envelope u and the wave field η are linked by the relation $\eta(x,t) = \mathrm{Re}\left\{ u(x,t)\exp i\left(\frac{1}{2}\omega_0 t - k_0 x\right)\right\}$ where $\mathrm{Re}\{z\}$ denotes the real part of the complex number z.

We use a numerical set-up very similar to the one of Adcock and Taylor (2009). We solve the NLSE using a fourth-order Runge–Kutta scheme in time and a pseudo-spectral method in space. We set a peak period of 12 s and use a time step of 0.14 s. We use 2^{12} Fourier modes and a spatial discretization of 10 m. All the simulations described here involve the following first steps:

- Take a linear GWP at focus with amplitude A_f^L and width $L_\mathrm{f}^\mathrm{L} = 5/k_0$ (we now add the L superscript to discriminate linear and non-linear quantities).
- Back-propagate the GWP according to linear evolution for 65 time periods.
- Use this back-propagated GWP as an initialization and solve the NLSE numerically until the GWP is focused. Identify the focusing time t_f when the spatial maximum of wave amplitude $\max_x(|u(x,t)|)$ reaches a maximum. Measure A_f^NL and L_f^NL, the amplitude and width of the wave packet at focus, by fitting a Gaussian to the modulus of the envelope $|u(x, t_\mathrm{f})|$.
- Perform a cubic spline interpolation of u in space down to 3.3 cm resolution to allow for a precise measurement of crest velocities.

4.3 Non-linear crest velocity profiles

From those simulations, we measure $\partial_x c_\mathrm{sp}|_{x=0}$ as in Sect. 3.3, with one modification to the method. This time, we measure the gradient only when one crest or one trough is in the middle of the wave packet such that $|x| \leq 4\lambda_0 \mathrm{d}t/(2T_p)$, giving 4 measurements of the gradient around the position $x = 0$ every time period. By averaging those 4 measurements of $\partial_x c_\mathrm{sp}$, we get a better estimate of $\partial_x c_\mathrm{sp}|_{x=0}$, and we avoid the oscillations

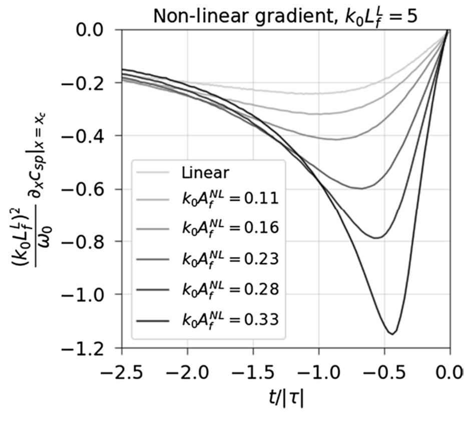

Fig. 6 Phase velocity gradient estimated from crest velocities of non-linear GWP, as a function of time. The non-linear evolution of one-dimensional GWP is approximated through numerical simulations of the NLSE

observed on the dotted lines in Fig. 4. Finally, to draw a complete profile of $\partial_x c_{sp}|_{x=0}$ over time, we repeat this procedure for different values of the constant and uniform phase.

Figure 6 shows specular velocity gradient (approximated from crest and trough velocity measurements) versus time for different degrees of non-linearity. Focusing is always set at $t = 0$. Non-linear effects amplify the focusing, leading to a higher ratio A_f^{NL}/A_f^L and thus a lower ratio L_f^{NL}/L_f^L. This strengthening of the spatial concentration of energy is consistent with the higher values of $\partial_x c_{sp}|_{x=0}$ measured in Fig. 6 for higher values $k_0A_f^{NL}$. The minimum of $\partial_x c_{sp}|_{x=0}$ is reached at $t = -|\tau|$ in the linear case, and it is reached for values of t closer to 0 for non-linear wave packets.

Numerical experiments for $k_0L_f^L = 10$ (not shown here) indicate that the shape of the non-dimensional profile $\frac{(k_0L_f^L)^2}{\omega_0}\partial_x c_{sp}|_{x=0}(t/|\tau|)$ depends on both $k_0A_f^L$ and $k_0L_f^L$ when higher-order non-linear effects are accounted for. As we have seen in Sect. 2, this profile was independent of $k_0A_f^L$ and $k_0L_f^L$ in the linear, narrow-banded case (low enough $k_0A_f^L$ and high enough $k_0L_f^L$).

We then evaluate the spatial profile of non-linear specular velocity at the time when the measured gradient $\partial_x c_{sp}|_{x=0}$ reaches a minimum value. This is plotted in Fig. 7. We see that the quasi-quadratic spatial profile of specular velocity from Eq. (6) is not valid for non-linear wave packets. For weakly non-linear GWP ($k_0A_f^{NL} \leq 0.16$), we see that the gradient at $x = 0$ is higher than in the linear case. For even higher values of $k_0A_f^{NL}$,

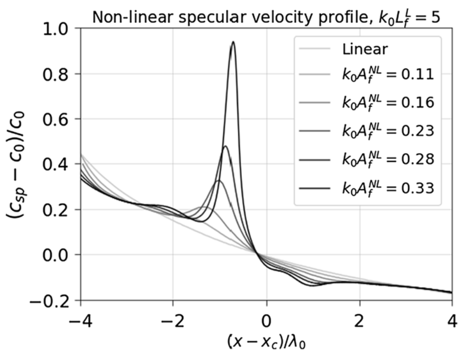

Fig. 7 Crest velocity of non-linear one-dimensional GWP, as a function of the space coordinate x. The profile is estimated at the time when the central gradient of phase velocity is the lowest (see Fig. 6). The non-linear evolution of one-dimensional GWP is approximated through numerical simulations of the NLSE

the spatial scale of variations of c_{sp} is too small to estimate its spatial profile based on measurements of crest velocities which are separated by one wavelength in space.

A theoretical study of phase velocities for weakly non-linear wave fields is given in section VII of Yuen and Lake (1982). Figures 43, 44 and 45 show the departure from the linear waves relation $c_{\mathrm{p}}(k) \propto k^{-1/2}$. For various spectra, the study shows that the phase velocity ratio $c_{\mathrm{p}}(k)/c_{\mathrm{p}}(k_0)$ is augmented for wave numbers both above and below the peak. In our focusing GWP case, shorter waves are in the front (downstream) and longer waves in the back (upstream), and both phase velocities will be augmented due to non-linear terms. To conclude this analysis, one would have to use information on the spatial distribution of wave numbers in a weakly non-linear focusing wave packet. However, the study of Yuen and Lake (1982) is a starting point for the theoretical evaluation of the behaviour shown in Fig. 7.

4.4 Applying the linear method to weakly non-linear GWPs

Our method relies on fitting $\partial_x c_{\mathrm{sp}}|_{x=0}$ as measured with crest and trough velocities to the simple profile of $\partial_x c_{\mathrm{sp}}|_{x=0}$ from Eq. (6). Section 4.3 shows that this profile is modified by higher-order non-linear effects. To apply our method, we also need to be able to measure $\partial_x c_{\mathrm{sp}}|_{x=0}$ based on crest velocity measurements, which can become impossible for very high values of $k_0 A_{\mathrm{f}}^{\mathrm{NL}}$. However, our method should still be applicable to weakly non-linear GWPs with a small enough amplitude. As we have seen in the previous section, higher-order non-linear effects on crest velocities seem to be small enough that our method is likely to provide consistent results when amplitudes are below the approximate threshold $k_0 A_{\mathrm{f}}^{\mathrm{NL}} \leq 0.16$.

Following this reasoning, we applied our method directly to estimate t_{f}, $A_{\mathrm{f}}^{\mathrm{NL}}/A^{\mathrm{NL}}(t)$ and $k_0 L_{\mathrm{f}}^{\mathrm{NL}}$ over non-linear wave packets with parameters $k_0 L_{\mathrm{f}}^{\mathrm{L}} = 5$ and $k_0 A_{\mathrm{f}}^{\mathrm{L}} = 0.13$ (which correspond to $k_0 A_{\mathrm{f}}^{\mathrm{NL}} = 0.16$ in our simulations of the NLSE). We repeated this estimation procedure over several wave packets with different values of the constant and uniform phase (40 values seemed enough to capture the variability of the results). We added only a minor change to our method. As in the previous section, we measured the specular velocity gradient only when a crest or trough was lying close enough to the centre ($|x| \leq 4\lambda_0 \mathrm{d}t/(2T_p)$). This provides fewer points for the estimation of the profile of $\partial_x c_{\mathrm{sp}}|_{x=0}$, but these points are of better quality, especially since the oscillations witnessed in Fig. 4 are strengthened by higher-order non-linear effects.

The results of this test are shown in Fig. 8. For values of $t_{\mathrm{end}} \leq -23\,T_p$, despite some variability in the results, most points converge to a satisfying estimation of t_{f} and very large overestimation of $A_{\mathrm{f}}^{\mathrm{NL}}/A^{\mathrm{NL}}(t_{\mathrm{end}})$ and underestimation of $k_0 L_{\mathrm{f}}^{\mathrm{NL}}$. This inaccuracy is much larger than the errors shown in Fig. 5. This shows that more points are needed for the linear method to provide acceptable results for weakly non-linear wave packets than are needed for linear wave packets. For $t_{\mathrm{end}} \geq -23\,T_p$, the method converges to a fairly good estimation of $A_{\mathrm{f}}^{\mathrm{NL}}/A^{\mathrm{NL}}(t_{\mathrm{end}})$ and $k_0 L_{\mathrm{f}}^{\mathrm{NL}}$, while t_{f} is underestimated by a few wave periods. These results indicate a certain stability of our method with respect to higher-order non-linear effects, such that consistent results can still be achieved for weakly non-linear wave packets if enough measurements of crest/trough velocities are provided.

However, as we have seen, our method is not applicable for higher-order non-linear wave packets above a certain amplitude threshold. This threshold depends on $k_0 L_{\mathrm{f}}^{\mathrm{L}}$, and our simulations show that for $k_0 L_{\mathrm{f}}^{\mathrm{L}} = 5$ it is around $k_0 A_{\mathrm{f}}^{\mathrm{NL}} \approx 0.16$. Precise determination of this threshold requires a more thorough examination. Note that the one-dimensional NLSE

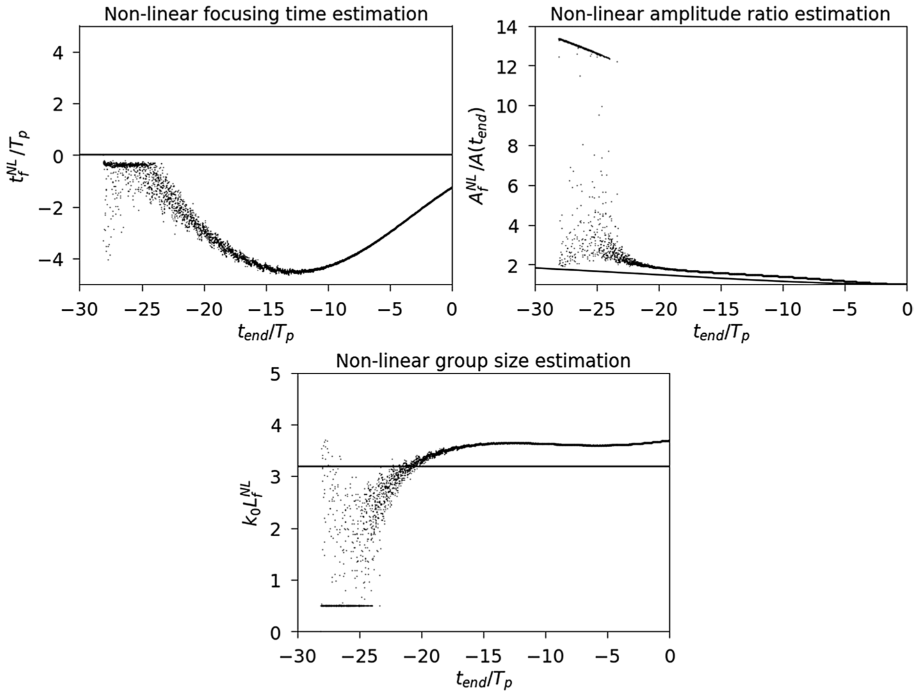

Fig. 8 Estimation of t_f, $A_f^{NL}/A^{NL}(t)$ and $k_0L_f^{NL}$ for non-linear wave packets with parameters $k_0A_f^L = 0.13$ and $k_0L_f^L = 5$ ($k_0L_f^{NL} \approx 3.2$). We use directly the method that was designed for linear wave packets, and we start observing crest velocities at $t_{start} \approx 30\,T_p$. True values (objectives) are in full black lines. Estimations are in black dots. Here, we present many estimations coming from simulations of GWPs with different values of the constant phase. The non-linear evolution of one-dimensional GWPs is approximated through numerical simulations of the NLSE

overestimates the increase in amplitude due to non-linear effects compared to more precise, higher-order evolution equations (Cousins and Sapsis 2015a). Therefore, one might under-estimate the threshold above which our method is not valid when using numerical simulations of the NLSE.

Above this threshold, generalization of our method is delicate and demands further study. A natural path would be to use analytical solutions for weakly non-linear GWPs such as those in Pizzo and Melville (2019). Another possibility would be to use a data-driven approach to learn how our method should be modified to take into account higher-order non-linear effects.

To investigate the consequences of such a threshold for real applications of our work in the ocean, we used the data analysed in Ruju et al. (2020) for several strong storms, with H_s values ranging from 5 to 20 m. With these data, we used the definition of design wave used for offshore structures. This design wave is the largest wave encountered out of 1000 waves, on average, and equals $1.86H_s$ for linear waves. While maintaining awareness of the limitations of such an approximation, we use it to find an order of magnitude of k_0A_{max}, the largest adimensional wave amplitude one would be likely to face in such storms. By doing so, we found values of k_0A_{max} mostly between 0.1 and 0.22, with a peak at 0.25. If the aforementioned threshold amounts to about 0.16, these values of k_0A_{max} indicate that our method could be used for many waves encountered in these storms, but it would need some modifications to be applicable to the steepest storm waves.

5 Conclusion

We have shown that the focusing time, the ratio of amplitude at focus to instantaneous amplitude and the width at focus of an isolated GWP can be inferred from the measurement of crest and trough velocities over time near the centre of the GWP. Our analytical derivations assume the linear dispersion of one-dimensional ocean surface gravity waves. The proposed methodology has been successfully tested over numerically simulated linear Gaussian wave packets with spectral widths of 0.15, 0.071 and 0.035, thereby proving relevant in terms of spectral width for dominant waves in storm-like conditions.

The linear approximation breaks down for real surface waves of finite amplitude. However, this work is still directly applicable to second-order non-linear wave packets. Analysing the crest/trough velocities of higher-order non-linear GWPs simulated numerically with the NLSE, we found that our method shows a certain stability to higher-order non-linear effects, and that it becomes inapplicable above a certain amplitude threshold. The evaluation of this threshold was not performed here and would require a thorough investigation, as it depends on the spectral width of the GWP. Above this threshold, our method could be generalized using either approximate analytical models for the evolution of higher-order non-linear wave packets, or data-driven tools from the machine learning community.

Still, in many situations the forecasting of rogue waves in moderate seas may be helpful. This is the case for marine operations at sea or transfer of workers from a ship to an offshore wind turbine. Our decision to rely solely on crest/trough velocities was motivated by both the sea surface reconstruction technique using lidar illumination and the combination of X-band radar images and stereo video cameras.

This study is a starting point to further investigate the applicability of our scheme to realistic random ocean surface gravity wave fields. First, the interactions between different wave packets were not considered here. Such interactions typically happen on larger timescales than wave packet contraction, and are therefore often put on the back burner. Secondly, the corrections to be added when taking directionality into account are an important concern. Directionality affects the higher-order non-linear changes to wave group contraction, mitigating the amplitude increase and inducing strong changes in the shape of the wave packet at focus (Adcock et al. 2012). Broadening the range of application of our approach is vital, as the combination of wave group interactions and directionality could be responsible for the formation of one of the largest waves measured at sea (Adcock et al. 2011).

The measurement of crest/trough velocities in short-crested seas is also obviously a technical challenge. However, to use our method on real wave fields from measurements, one could filter out the high frequencies that contain both instrumental noise, high-frequency wind waves and second-order non-linearities. Since none of those are either responsible for or interacting with the long-wave crest velocity behaviour described in this study, the methodology should be applicable to the filtered measurements. Note also that wavelength could be used in the same way as crest velocities, as wavelength is proportional to the square of the velocity for long-crested waves, and wavelength can be measured approximately from zero down- and up-crossings, which is less sensitive to high-frequency noise. In this case, the analog of the gradient of crest velocity would be the gradient of wavelength. This suggested counterpart methodology could be investigated in further research, but is yet likely to yield other unexpected limitations.

Wave energy dissipation and wind energy input are usually considered negligible for the forecast horizons considered here, which are inferior to approximately 60 T_p. In fact, integrated parameters of the wave spectrum such as H_s are usually estimated from data collected during approximately 100 T_p (e.g. Mitsuyasu et al. 1975). Forecast above this threshold of $\approx 60\,T_p$ is not considered for two reasons. First, instrumental noise makes it hard to forecast individual waves more than 100 T_p ahead (see the predictability horizon introduced by Alam 2014). Second, it is hard to imagine measuring waves more than 60 peak wavelength (≈10 km) away from a ship or an offshore platform using on-board measurement devices. For now, the spatial horizon of X-band marine radars is limited to ≈2.5 km and is even smaller for stereo video cameras (Benetazzo et al. 2018).

Numerical and wave tank experiments of Kharif et al. (2008) showed that for large amplitude waves, wind–wave interactions can both weakly increase the maximum wave amplitude and noticeably shift the focusing point downstream. This further influence of wind would have to be parameterized in addition to the proposed methodology.

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Data accessibility All the data and python scripts used to produce the figures of this article are publicly available in the following GitHub repository: https://github.com/PaulPlatzer/Wave_Focusing_Crest_Velocities.

Compliance with ethical standards

Conflict of interest The authors declare that they have no conflict of interest.

Appendix: Derivation of Ψ_G^a

Let us write

$$\Omega(k) = \omega_0 + c_g(k-k_0) + \frac{c^*}{2k_0}(k-k_0)^2 + \text{h.o.t.}, \quad \Phi(k) = \phi_0 - x_0(k-k_0) + \frac{S}{2}(k-k_0)^2 + \text{h.o.t.},$$

where $x_0 = -\frac{\mathrm{d}\Phi}{\mathrm{d}k}(k_0)$ and $S = \frac{\mathrm{d}^2\Phi}{\mathrm{d}k^2}(k_0)$. We can then write Ψ_G as

$$\Psi_G = \frac{A_\mathrm{f} L_\mathrm{f}}{\sqrt{2\pi}} e^{i(k_0 x - \omega_0 t + \phi_0)} \int \exp\big(f(k-k_0) + \text{h.o.t.}\big) \mathrm{d}(k-k_0),$$

where

$$\forall K,\ f(K) = -0.5\left[L_\mathrm{f}^2 + i\left(\frac{tc^*}{k_0} - S\right)\right]K^2 + i\big[x - c_g t - x_0\big]K.$$

Since $L_\mathrm{f} k_0 \gg 1$, we can neglect higher-order terms in the exponential and integrate with respect to $k - k_0$. This is a specific case of Laplace's method. Rearranging, this gives Eq. (2), but with

$$L^2(t) = L_\mathrm{f}^2 + \left(\frac{\frac{c^* t}{k_0} - S}{L_\mathrm{f}} \right)^2, \quad \phi(x,t) = -\arctan \left(\frac{\frac{c^* t}{k_0} - S}{2L_\mathrm{f}^2} \right) + \frac{(x - x_c(t))^2}{2L^2 L_\mathrm{f}^2} \left(\frac{c^* t}{k_0} - S \right),$$

and with $x_c(t) = x_0 + c_g t$. Setting $t \to t - Sk_0/c^*$, and $x \to x - x_0 + Sk_0 c_g/c^*$, we recover the expressions of Sect. 2.2.

Using the work of Olver (1968), one can show that Ψ_G^a gives an approximation of Ψ_G/A_f that is valid up to the second order in $1/(k_0 L_f)$, the next order being the fourth one. It is also clear from the work of Olver (1968) that obtaining higher-order approximations of the right-hand side of Eq. (1) would require information on the 3rd and higher-order derivatives of $\Phi(k)$ at k_0 that could not be set to zero by changing the origin of the time and space axis.

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